

Evaluating the Quality of Counterfactual Explanations in Multivariate Time-Series

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Abstract

Counterfactual explanations (CFEs) offer a promising approach for understanding personalized psychological processes captured through Ecological Momentary Assessment (EMA). In particular, generating CFEs for time points associated with mental health deterioration can help identify alternative changes that might prevent such outcomes. To assess the quality of generated CFEs on time-series data, we propose a 3-level framework: per explanation (based on feature changes, proximity, and model confidence), per time point (based on structure and diversity through clustering), and per individual temporal dynamics. Applying this framework to a real-world time-series EMA dataset, we demonstrate how we assess the validity and interpretability of large volumes of CFEs.

CCS Concepts

• **Computing methodologies** → **Temporal reasoning**; • **Applied computing** → **Health care information systems**; • **Information systems** → **Decision support systems**.

Keywords

Counterfactual Explanations; Time-series Explanations; Explainable AI; Ecological Momentary Assessment; Mental Health

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1 Introduction

Mental disorders are characterized by high relapse rates after treatment, partly due to the misconception that they arise from a set of independent symptoms, which leads to inaccurate diagnoses and suboptimal treatment decisions [2]. However, mental disorders are complex systems of dynamically interacting symptoms and risk

factors, such as emotions, behaviors, subjective experiences, and contextual influences, that fluctuate over time [1].

Given the complexity of mental disorders, there is a clear need for continuous and context-aware monitoring of individuals' mental states in daily life [15]. To address this, Ecological Momentary Assessment (EMA) offers a powerful approach to collect repeated and real-time data on various patients' psychological variables. Specifically, through digital questionnaires, EMA can monitor fluctuations in emotions, behavior, and context over time in an individual's natural environment. This high-resolution temporal data provides a rich foundation for modeling changes in mental state and predicting clinically relevant outcomes [12]. For instance, predictive models applied to EMA data can be used to forecast the next time point of an EMA variable, such as detecting moments of increasing negative affect (NA).

Building on such predictive models, a key next step is to not only predict deterioration but also to understand why and how it might have been prevented. Understanding the mechanisms behind predicted psychological changes is crucial for designing interventions but also sending timely alerts or suggestions. To facilitate this, counterfactual explanations (CFEs) provide a promising framework for identifying actionable changes that could have prevented predicted deterioration [18]. By uncovering minimal and plausible changes to input features that could have changed a model's prediction, such as preventing a future increase in negative affect (NA), CFEs reveal potential intervention points tailored to an individual's information. This can support clinicians in developing more personalized interventions or timely suggestions.

However, even when generating CFEs for a single instance or time point, the produced explanations are often numerous, redundant, and difficult to interpret. Although all CFEs succeed in flipping the model's prediction, what constitutes a meaningful or feasible change can vary significantly, and such considerations are crucial in practice. Thus, in this paper, we propose a framework to analyze and evaluate CFEs in temporally structured data. Specifically, we propose to assess CFEs at three levels: (1) per explanation, by examining feature changes, proximity to the original instance, and prediction confidence, (2) per time point, by analyzing the structure and coherence of CFEs using clustering and silhouette scores, and (3) per individual, by interpretively exploring the temporal dynamics of changes suggested across time. Overall, this framework aims to enhance CFEs validity and interpretability at the individual level



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in real-world EMA applications. While this work is grounded in the context of mental health monitoring, the proposed approach is generalizable and can be extended to other domains involving multivariate time-series data where personalized and interpretable decision support is needed.

2 Related Work

Explainability for time series data remains an open challenge, particularly in health and behavioural monitoring domains. The complexity of temporal and multivariate data, such as that collected through EMA, requires tailored approaches to enhance model transparency. While attribution methods, e.g., SHAP and LIME [9, 14], have been adapted for time series, they often violate assumptions such as feature independence, limiting their effectiveness. Attention mechanisms offer built-in interpretability but may not align with human reasoning across diverse application domains [6]. Shapelet-based models show promise by identifying discriminative subsequences, though they can be limited in multivariate settings. Growing interest in eXplainable AI (XAI) for ubiquitous and wearable computing has highlighted the need for methods tailored to multi-modal and temporally dynamic data [4].

CFEs for time series aim to identify minimal and plausible changes that alter a model's prediction. They are particularly compelling in human-centered domains because they align with how people reason and provide explanations, and support actionable, personalized insights [11, 13]. Generating high-quality CFEs is challenging, as it requires balancing multiple objectives—such as proximity (remaining close to the original instance), sparsity (changing as few features as possible), plausibility (ensuring changes are realistic and within the data distribution), and predictive validity (successfully flipping the model's prediction)—within a high-dimensional and often complex search space. For example, the counterfactual generator BayCon addresses these challenges through probabilistic feature sampling with Bayesian optimization to efficiently produce diverse and plausible counterfactuals [16]. While methods, such as SG-CF and TSEvo [5, 8] attempt to improve plausibility and interpretability for time-series data, challenges remain around redundancy, conflicting feature changes, and lack of temporal coherence. Our work builds on top of BayCon and introduces novel tools for summarizing and tracking CFEs across time, enabling a more structured and temporally aware approach to personalized intervention design.

3 Problem Formulation

EMA Dataset

The examined dataset is a real-world EMA dataset collected from university students in the Netherlands [10, 15]. It includes time-series data from 187 individuals, capturing 12 psychopathology-related variables, such as positive and negative affect, cravings (for food or substances), stress, control, and social context. These variables were rated on a 7-point Likert scale and recorded 8 times per day for 4 weeks, resulting in an average of approximately 150 time points per individual.

Predictive Task: Predicting Negative Affect (NA)

The goal of the predictive modeling task is to forecast negative affect (NA) at the next time point based on the individual's current state. Specifically, given all EMA variables at time point t , the model

is trained to predict the level of NA at time point $t + 1$. For this task, the output variable is dichotomized based on each individual's mean value, with a value of 1 indicating high NA (deterioration) and 0 indicating low NA.

Predictive Model

Although counterfactual explanation methods are generally model-agnostic, they heavily depend on the performance and internal reasoning of the input predictive model. CFEs rely on the learned decision boundaries of the model to propose alternative scenarios that would change the predicted outcome. Therefore, generating meaningful and reliable CFEs requires a well-performing predictive model.

In this case, an XGBoost model [3] is applied independently to each individual's data to capture personalized EMA patterns [12]. The model achieves an average AUC score of 0.60 ($std = 0.05$) across participants, evaluated on the test set after applying a 70/30 sequential split of each individual's time series data. Due to the observed variability in performance between individuals, we focus on analyzing a single individual within the training setting. For the examined individual, the most important variables influencing the prediction of high NA are the *Somatic negative affect* and being *In control*, followed by *enjoying social activities*, which also appear in CFEs discussed later.

Generating Counterfactual Explanations

Generating high-quality CFEs in multivariate temporal data involves balancing multiple objectives—such as proximity, sparsity and plausibility—within a complex search space [7]. Achieving these objectives requires advanced optimization, particularly when aiming to produce informative and actionable explanations in human-centered domains.

We build on BayCon, a model-agnostic counterfactual generator that leverages probabilistic feature sampling and Bayesian optimization to generate diverse and realistic explanations while respecting constraints like immutability and feature distributions. BayCon evaluates candidates using an objective function that integrates feature similarity (via Gower distance [17]), output similarity, and sparsity. It also uses a surrogate model (e.g., XGBoost) to approximate the predictions of the original black-box model.

In our work, we extend BayCon's functionalities to address temporal dynamics, an aspect largely underexplored in the counterfactual literature. Rather than generating CFEs in isolation, we analyze their evolution across time, enabling the identification of trends, shifts, and potential intervention windows that are consistent and interpretable over time.

4 CFEs Evaluation

The following 3-step evaluation framework focuses on a single individual selected based on an adequate number of data points, satisfactory test performance and necessary variability in features. For this individual, the model predicts Negative Affect (NA=1) at 38 time points. The number of CFEs varies considerably across instances, ranging from zero to several hundred (e.g., 700–800). A key factor influencing this variability is the distance of the examined instance to the training data distribution. Instances closer to the well-represented regions of the data tend to yield a larger number of CFEs, as more plausible alternatives can be generated in their

local neighborhood. Consequently, instances that lie further from the majority of the training data, such as outliers, may result in few or even no valid CFEs, as there are limited alternatives within the valid data distribution that could alter the prediction. Thus, evaluating the quality or meaningfulness of each explanation is a crucial step towards identifying the most plausible changes.

4.1 Instance-based Evaluation per Individual CFEs

At the first level, we examine all generated CFEs for a single instance (or time point) associated with a predicted psychological deterioration, i.e., high NA. This allows us to assess the quality of each CFE individually while the collective assessment of all CFEs for a given instance provides an overview of the counterfactual landscape surrounding that prediction. In this setting, three key aspects are taken into account in CFEs quality assessment:

- Sparsity, the number of features changed in each CFE (*Feat*)
- Proximity, how close each CFE is to the instance of reference (*L1*)
- Model confidence, what is the probability of each CFE to reverse the outcome (*Prob*)

All three quantities are normalized (separately per instance) to the range $[0, 1]$ to ensure comparability and the final ranking score is defined in Equation 1. A higher score indicates a better-quality CFE, e.g., one with minimal changes, small distance to the original input, and high probability of reversing the output. The distribution of ranking scores for all generated CFEs across all 38 time points is presented in Figure 1. Overall, a wide range of scores within each instance is observed, highlighting a significant variability in the quality of the CFEs. This large variability reflects the generation of multiple directions for interventions. Although the average scores range around 60–70 in all instances, there are a lot of low-scored outlier CFEs. Thus, it is important to evaluate and filter the extracted CFEs regarding their feasibility, interpretability and practical utility.

$$\text{RankingScore} = \left(1 - \frac{\text{Feat} + L1 + (1 - \text{Prob})}{3} \right) \times 100 \quad (1)$$

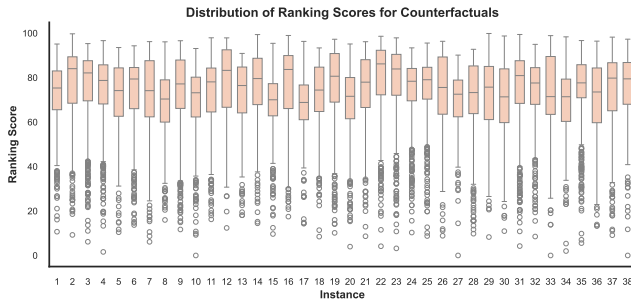


Figure 1: Distribution of ranking scores across all generated CFEs per instance.

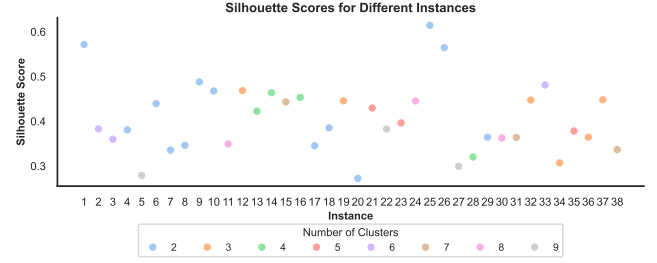


Figure 2: Maximum Silhouette scores after clustering all CFEs per instance, colored by the number of extracted clusters.

4.2 Instance-based Evaluation through Clustering

Moreover, we evaluate instance-based CFEs at a more abstract level by applying clustering to uncover the underlying structure of the CFEs. Grouping CFEs based on their feature values enables us to assess the diversity and coherence of the generated alternative suggestions, while also summarizing the large volume of extracted CFEs. To quantify this structure, we use the silhouette score, which captures the similarity of each CFE cluster relative to its most similar cluster.

In particular, for each instance, all CFEs are used as input to k-Means clustering, exploring a range of cluster numbers from 2 to 10. The optimal number of clusters is selected based on the highest silhouette score, ensuring that the resulting clusters are both internally coherent and well-separated. The highest silhouette score across all instances is shown in Figure 2, colored by the number of extracted clusters. Higher silhouette values indicate that the model offers well-separated and consistent alternative suggestions, whereas lower scores reflect highly diverse CFEs. On average, the silhouette score was 0.41, with 17 instances exceeding a score of 0.5, suggesting that in almost half of the cases the CFEs form clear clusters. Interestingly, both high and low scores occur across different numbers of clusters. Overall, clustering was performed to evaluate whether CFEs could be successfully summarized into distinct and interpretable changes.

4.3 Interpretability of CFEs per Individual

Beyond quantitative evaluation, the next step is to qualitatively assess the interpretability of the CFEs, exploring the consistency of their temporal patterns and their plausibility. To investigate this, we analyze how CFEs vary across all consecutive instances with a prediction of high NA. This facilitates examining whether the model tends to suggest similar actions at different time points or different ones depending on how EMA evolves.

For each time point associated with a predicted high NA, we examine the frequency (in percentages) and direction of feature changes as illustrated in Figure 3. The figure shows bar plots where changes in feature values are stacked above or below the zero line (baseline): bars above zero indicate positive changes, while those below indicate negative changes. While the total height of the bars is not directly meaningful, the relative height of each bar reflects the average frequency that this feature is changed across all CFEs.

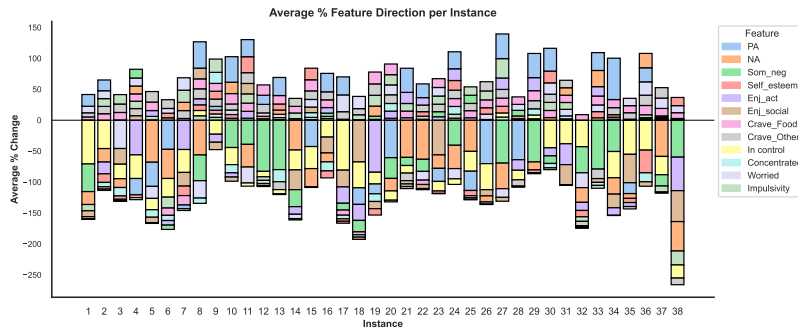


Figure 3: Stacked bar plot of average directional CFEs change per feature (indicated by a different color) across all instances.

Within each direction (positive or negative), features are ordered from the highest to the lowest average percentage of change.

Interestingly, recurring spikes are observed in variables, such as *In control*, *PA*, *NA* and *Somatic NA*, which are also among the model’s most important variables, highlighting their consistent involvement in the model’s suggestions. Examining the direction of these features reveals that *In control*, *NA* and *Somatic NA* are mostly connected to negative changes. Moreover, we can also notice more details about features with changing directions over time. For instance, *PA* changes between positive and negative direction, while in most cases it shows a positive change, often emerging as the feature with the highest overall positive magnitude.

From a plausibility perspective, many of the suggested changes align with psychological theory. While decreasing *NA* and *Somatic Negative Affect* aligns with common treatment strategies, decreasing *In control* is not so intuitive. Such cases reflect the high dependence on the underlying predictive model along with the lack of any data constraints. Therefore, involving domain experts is essential to assess the plausibility of the suggested changes but also how feasible it is to intervene on these.

5 Discussion - Conclusions

In this work, we proposed a multi-level framework to evaluate counterfactual explanations in the context of EMA data. Specifically, the focus was on generating and evaluating CFEs for time points or instances predicted as high *Negative Affect (NA)*, with the goal to identify potential changes that could prevent such deterioration in an individual’s mental state. Given the large number of CFEs extracted for each instance, there was a need to assess CFEs on three levels: (1) per CFE, (2) per instance, and (3) collectively per individual.

First, we evaluated the quality of generated CFEs individually based on three key aspects: sparsity, proximity to the training instance and model probability. This ranking score revealed wide variability across CFEs, with many of the suggested changes to be outliers, marking them as potentially not plausible or complex suggestions. Thus, CFEs further evaluation and filtering are necessary next steps, potentially including adjustments to the ranking score to more heavily penalize certain aspects.

This was followed by the assessment at the instance or time point level, checking the structure and diversity of the possible alternatives through clustering. While the majority of instances showed a moderate structure and separation across suggested changes, several instances showed poorly separated or overlapping CFEs. This level of evaluation adds insight into the reliability of counterfactual suggestions and facilitates summarizing them and identifying the most interpretable and actionable cases.

Moreover, it is important to take into account the temporal dynamics of EMA data, investigating changes over time. This offers great insights into the most consistent directional changes across time points, which potentially indicate stable intervention signals. In future work, such temporal patterns should be further investigated in order to identify recurring motifs as well as analyze similarities between different individuals. This could facilitate grouping similar behavioral patterns and discovering more generalized intervention strategies suitable for a broader population. Beyond the EMA context, the proposed framework is broadly applicable to other time-series domains where interpretability and personalized decision support are essential, such as physiological monitoring, user modeling, or behavior forecasting. This can offer a structured approach to evaluate counterfactual explanations in multi-level sequential settings.

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